Deposition and Characterization of Organic Mixed Ionic and Electronic Conductor

Alessandro Ciavatta, Elisa Damiani, Riccardo Gaeta

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Abstract

An ITO (Indium Tin Oxide) electrode is coated with a PEDOT:PSS thin layer through an electrodeposition process. The total charge used in the process allows to determine the total deposited volume to then obtain the layer thickness l and total theoretical capacitance C_V: $l = 168 \pm 8\,\text{nm}$, $C_V = 990 \pm 50\,\mu\text{F}$. By fitting impedance spectroscopy data of the coated electrode with an RC series circuit these values are acquired: $R_{coat} = (27.3 \pm 0.2)\,\Omega$ and $C_V = (831 \pm 11)\,\mu\text{F}$. The impedance spectroscopy of the bare ITO electrode immersed in a PBS solution is itself fitted with and RC series resulting in: $R_{bare} = (30.8 \pm 0.9)\,\Omega$ and $C_{bare} = (18.5 \pm 0.5)\,\mu\text{F}$. A further study on the ITO/PBS interface is performed by means of a Mott-Schottky analysis, which provides a dopant density of $N_D = (7.5 \pm 0.8) \times 10^{21}\,\text{cm}^{-3}$ and a flat band potential of $V_{fb} = (-0.76 \pm 0.05)\,\text{V}$.

Introduction

The aim of this work is to produce and characterize a thin film of PEDOT:PSS, an organic mixed ionic and electronic conductor (OMIEC).

OMIECs are nanoscale phase separated materials, characterized by crystalline electronic conductor domains in an amorphous ionic conducting matrix. In the past few decades, they have emerged as ideal materials for many applications, from energy storage to health technologies, thanks to their enhanced charge storage and coupled transport properties [1].

PEDOT:PSS is one of such OMIECs as it's a polymer composite made out of polymeric chains of EDOT (3,4-ethylenedioxythiophene) and PSS (poly(sodium 4-styrenesulfonate)). Its chemical structure is shown in Fig.(1).

A controlled way to synthetize PEDOT:PSS is through electrodeposition: an electrode is put in a solution containing EDOT monomers, an applied positive potential rips an electron off the monomers creating positive radicals that then bond together at the electrode surface forming dimers. For each created dimer two protons and two electrons are released. The electrons will be captured by the electrode, while the protons are released in the solution. Larger and larger molecules lose electrons and aggregate to form longer and longer chains attached to the electrode surface [3].

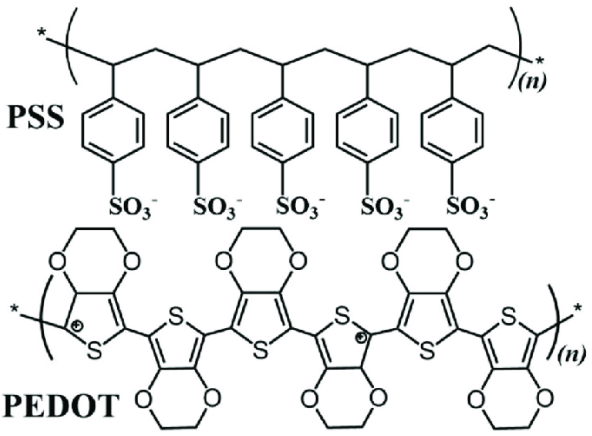


Figure 1: Chemical structure of PEDOT:PSS [2]

PEDOT is an organic semiconductor with a bandgap of around 2 eV. By removing an electron from the chain PEDOT is oxidised (p-doped state) and it becomes conductive, with a hole mobility of around $1\,\text{cm}^2/\text{Vs}$, thanks to the charge delocalization provided by the C=C π orbitals [3]. The positive charge of the oxidised

PEDOT is counterbalanced by the co-deposited PSS as it easily loses its Na^+ ions (usually in exchange for hydrogen) and thus can readily harbour negative charges (Fig.(1)).

This process allows to electrically switch the doping of the PEDOT:PSS if it immersed in a solution containing positive ions. If the PEDOT:PSS is positively biased (in relation to the solution) it will become doped, as electrons are removed from the polymer, while if it is negatively biased the material will accept electrons becoming undoped and the negative PSS charge will be counterbalanced by a sodium ion that flows in the polymer composite matrix. The PEDOT:PSS in the undoped state is much less conductive as the carriers are ionic in nature and thus much less mobile ($\sim 0.001\,\mathrm{cm}^2/\mathrm{Vs}$).

Furthermore the positive charge in the doped state introduces bond length and angle distortions as well as longer range dipole interactions in the polymeric chains. This creates polaron quasiparticles (or even bipolarons) with their associated polaronic energy states that sit inside the undoped band gap. This changes the material's optical properties as the band gap is reduced shifting the optical absorption peak energy in the infrared region: PEDOT:PSS becomes roughly transparent. These properties are summarized in Fig.(2) below.

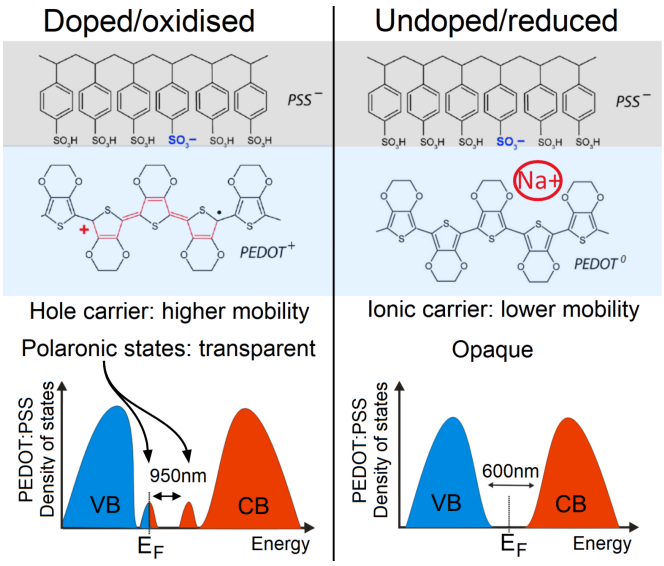


Figure 2: PEDOT:PSS doped and undoped configurations with their properties

The electrodeposition of PEDOT:PSS can be achieved by cyclic voltammetry. Cyclic voltammetry or cyclovoltammetry (CV), in a three-electrode potentiostat, is performed by linearly ramping the potential of the working electrode at a constant rate forward and backwards inside a potential range, while the current flowing in or out the working electrode is measured.

The measurement of this current allows to determine the amount of charge Q used during the polymerization process. Knowing the surface area A of the electrode, the total charge can be used to estimate the thickness of the deposited layer by using the first Faraday's law of electrolysis (the mass of the deposited species is directly proportional to the used charge):

$$l = \frac{\text{Volume}}{A} = \frac{MQ}{\rho e N_A A} \tag{1}$$

where M and ρ are the molecular weight and density of EDOT, respectively, e is the elementary charge and N_A is Avogadro's number.

In this work, PEDOT:PSS is deposited on a ITO (Indium Tin Oxide) electrode. To electrically analyze the bare ITO electrode and the PEDOT:PSS coated electrode their equivalent circuit must be understood. The bare ITO electrode is characterized in a PBS (phosphate-buffered saline) solution.

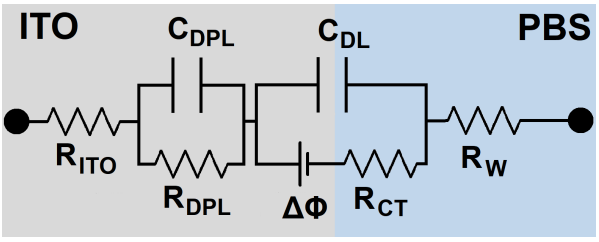


Figure 3: Equivalent circuit of ITO/PBS interface. $\Delta\Phi$ is the Nernst potential

The whole ITO/PBS interface is a semiconductor/liquid interface that can be modelled by a circuit, shown in Fig.(3), containing two capacitances: the capacitance associated to the depletion region C_{DPL} and the diffuse double layer capacitance C_{DL} that is due to the accumulation of charges at the interface. This circuit, with some approximations, can be simplified.

The charge transfer across a solid/liquid interfaces is due to electrochemical processes (such as electrolysis), which need to overcome a potential barrier to be activated. So for small voltages ($|V| < 1.23\,\mathrm{V}$) no charge transfer happens ($R_{CT} \to \infty$).

The ITO/PBS interface is akin to a metal/semiconductor junction and, if the voltage applied is small enough, its depletion layer can be modelled as an RC parallel circuit with a very large R_{DPL} leakage resistance (MΩ).

These two considerations allow us to reduce the whole circuit to an RC series circuit with the two capacitance C_{DPL} and C_{DL} in series.

The analysis of which capacitance dominates follows. The specific capacitance C_{DL}/A [F/cm^2] is given by:

$$\frac{C_{DL}}{A} = \frac{\epsilon\epsilon_0}{\lambda}$$

where ϵ_0 is the vacuum dielectric constant, ϵ is the relative dielectric constant of water ($\sim$80) and λ is the Debye screening length that is related to the mean thickness of the double layer. An estimation of λ is given by the Poisson-Boltzmann equation for a planar interface in the context of diffuse double layer theory, which takes into account both the electrostatics and the thermal motion of the ions:

$$\frac{\partial^2 E(x)}{\partial x^2} = -\frac{\rho}{\epsilon\epsilon_0} = \frac{ce}{\epsilon\epsilon_0}\left(e^{\frac{eE(x)}{kT}} - e^{-\frac{eE(x)}{kT}}\right)$$

whose solution (using $E(x) = E_0 e^{-x/\lambda}$) gives

$$\lambda = \sqrt{\frac{\epsilon\epsilon_0 kT}{2ce^2}}$$

where k is the Boltzmann constant, T is the temperature and c is the ions concentration.

Finally, the expression of the specific capacitance becomes:

$$\frac{C_{DL}}{A} = \sqrt{\frac{2\epsilon\epsilon_0 ce^2}{kT}} \qquad (2)$$

The depletion layer has itself an associated Debye screening length but this is usually much greater than λ_{DL}, meaning that $C_{DL} \gg C_{DPL}$ and therefore the total capacitance can be approximated by the depletion layer capacitance:

$$\frac{1}{C_{bare}} = \frac{1}{C_{DPL}} + \frac{1}{C_{DL}} \sim \frac{1}{C_{DPL}} \qquad (3)$$

The behaviour of C_{DPL} vs V can then be used to perform a Mott-Schottky analysis. A Mott-Schottky analysis consists of measuring the capacitance at different offset potentials to determine the dopant density N_D and the flat band potential V_{fb} of ITO by using the Mott-Schottky equation:

$$\frac{1}{C_{DPL}^2} = \frac{2}{eN_D\epsilon\epsilon_0 A^2}(V - V_{fb} - V_T) \qquad (4)$$

where A is the electrode surface, $\epsilon = 3.2$ is the relative dielectric constant of ITO [4] and V_T is the thermal voltage ($\sim$26 mV).

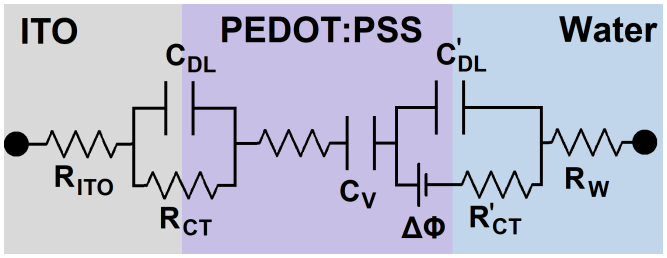


Figure 4: Equivalent circuit of ITO/PEDOT:PSS/Water interface

The equivalent circuit of the ITO/PEDOT:PSS/Water interface is more complex and is shown in Fig.(4).

Because of the band structure of ITO and PEDOT:PSS the depletion layer is absent [5] (the interface behaves like an ohmic contact) and the charge transport is facilitated by the hole carriers present in the PEDOT (R_{CT} small).

R'_{CT} is also small because the processes responsible for the charge transfer, are favored by the electrostatic interactions of PEDOT:PSS: the energy barrier is significantly lowered.

This circuit can thus be also reasonably approximated with an RC series circuit because all the resistances in parallel are small and so C_{DL} and C'_{DL} can be ignored. The only relevant capacity is the volumetric capacitance C_V. Thanks to its microscopic structure, PEDOT:PSS can store a large number of charges that grants it

a high volumetric capacity ($39\,\mathrm{F/cm^3}$ [6]) that must be taken into consideration even for very small layer thickness.

Materials and methods

A transparent ITO electrode is coated by electrodeposition with a nanometric thin film of conducting polymer PEDOT:PSS. The electrical properties of the ITO/PBS and the ITO/PEDOT:PSS/Water interfaces are characterized by impedance spectroscopy. Both of these techniques can be accomplished using a potentiostat with a three electrode setup. The potentiostat used is an Autolab PGSTAT 204 [7] and in Fig.(5) its schematics are shown. The reference electrode (RE) is connected to a feedback operational amplifier through a buffer, so that the current will flow only between the working electrode (WE) and the counter electrode (CE), made of platinum, a noble metal resistant to corrosion. In this work the RE is a saturated calomel electrode (SCE) made out of glass, with a small porous frit that separates the saturated solution of potassium chloride in water, mercury(I) chloride (Hg_2Cl_2) and elemental mercury. The RE has a defined electrode potential ($E = +0.241\,\mathrm{V}$ saturated) that allows the instrument to determine precisely the potential in the liquid.

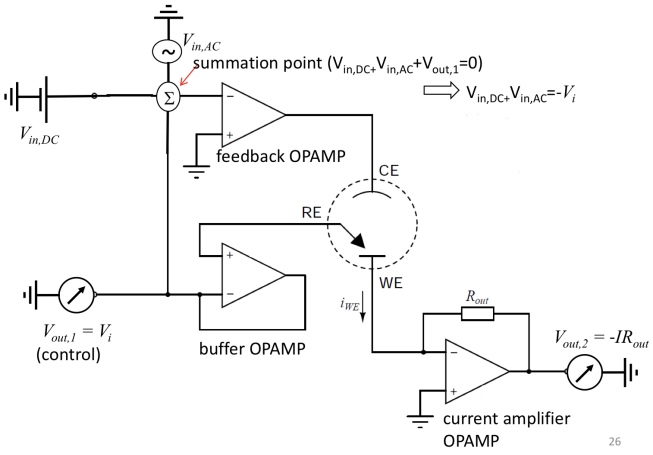


Figure 5: Schematic representation of the potentiostat with three electrode geometry

First of all, the ITO (Indium Tin Oxide) WE is prepared. It is a glass slide covered on one side with ITO. A piece of Teflon tape is used to expose an active area of $(1.50 \pm 0.08)\,\mathrm{cm^2}$. Silver paste is used to cover the ITO in order to make a good contact with the electrode holder.

The three electrodes are submerged in a beaker containing 50 mL phosphate-buffered saline (PBS) solution (containing, alongside a buffer, 0.1 M NaCl to achieve good ionic conductivity) and are connected via banana cables to the potentiostat. Using the Nova 2.0 software, a FRA (Frequency Response Analysis) spectroscopy is performed, with a frequency span of 0.1-10^5 Hz, an AC amplitude of 10 mVrms at 0 V DC offset in order to obtain the impedance spectrum of the ITO electrode. The resulting Bode plot is fitted by the software with a RC series equivalent circuit.

After identifying the frequency region in which ITO's impedance is mostly capacitive (< 10 Hz) and using a cyclovoltammetry scan (from 0 V to 1.5 V and then back to -1 V) to find the voltage region where no electrolysis occurs, the Mott-Schottky analysis for the ITO/PBS interface is carried out.

After the ITO electric characterization, the beaker is emptied and the instruments washed. The electropolymerization solution is prepared for the PEDOT:PSS electrodeposition. First, 25 mL of deionized water are poured in a the beaker. In order to reach the desired concentration of 0.1 mmol/L of PSS and 10 mmol/L of EDOT, 0.35 g of PSS are weighed on a digital scale and 0.053 mL of EDOT are measured with a micropipette. More deionized water is added in the end to reach a total volume of 50 mL.

The electrodes are put in the electropolimerization solution and the potentiostat performs a cyclic voltammetry, while the solution is mixed by a magnetic stirrer to facilitate mass transport. The potential is swept between 0 V and 1.5 V, with a scan rate of 0.1 V/s. This method synthesizes PEDOT:PSS that drops out of solution sticking on the electrode. Only one voltammetry cycle is performed.

An impedance spectroscopy of the ITO/PEDOT:PSS/Water interface is then performed and fitted with an RC series circuit. A Mott-Schottky analysis of this interface is not carried out.

Results and discussion

Impedance spectroscopy of ITO/PBS and ITO/PEDOT:PSS/Water

In Fig.(6) is shown the bode plot for the bare ITO electrode and for the ITO PEDOT:PSS coated electrode. The impedance of the coated electrode is mostly resistive over a wide frequency range, from 10 Hz to 10^5 Hz: the phase shift is close to 0°. The impedance of the bare ITO, instead, remains capacitive in a wider range (up to ∼30 Hz).

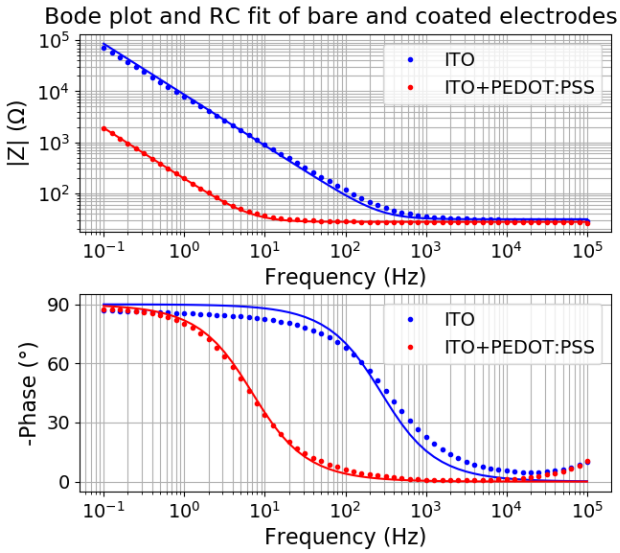


Figure 6: Frequency response of ITO/PBS and ITO/PEDOT:PSS/Water interfaces. The data are fitted with a RC series equivalent circuit for both samples

Since these two samples are studied in different environments, ITO is in a PBS solution, while the ions in the solution with the PEDOT:PSS sample are mostly provided by the PSS, comparing their behaviour is not entirely unquestionable. Nonetheless after the PEDOT:PSS deposition, the overall observed impedance of the electrode is reduced of around two orders of magnitude with the same steady frequency drop off of ITO. This increase in conductivity can be associated to the ability of PEDOT to facilitate both electronic and ionic transport and to the increase in the surface roughness of the electrode due to the coating [8].

The obtained results from the fit are shown in Tab.(1) below.

Bare ITO	$R_{bare} = (30.8 \pm 0.9)\,\Omega$ $C_{bare} = (18.5 \pm 0.5)\,\mu F$
Coated ITO+ PEDOT:PSS	$R_{coat} = (27.3 \pm 0.2)\,\Omega$ $C_V = (831 \pm 11)\,\mu F$

Table 1: RC series fit results

The fitted capacity of the ITO/PBS interface can be approximated by the capacitance of the depletion layer C_{DPL} since the double layer capacity $C_{DL} \gg C_{DPL}$. To establish this the Poisson-Boltzmann theory is used to calculate the theoretical C_{DL}.

C_{DL} estimation from Poisson-Boltzmann theory

The specific capacitance is calculated from Eq.(2) reported again here:

$$\frac{C_{DL}}{A} = \sqrt{\frac{2\epsilon\epsilon_0 c e^2}{kT}}$$

considering the relative dielectric constant of water ($\epsilon = 80$), a temperature T of 298 K and the molar concentration for the ions in the PBS solution c of 0.1 mol/L; multiplying by the area of the electrode ($1.50 \pm 0.08\,cm^2$), the value obtained for the diffuse double layer capacitance is $C_{DL} = (109 \pm 4)\,\mu F$. This value is indeed much greater than the measured value from the frequency response ($C_{bare} = (18.5 \pm 0.5)\,\mu F$). Using Eq.(3) the approximate value of C_{DPL} at 0 V can be extracted resulting in $C_{DPL} = (22.8 \pm 0.6)\,\mu F$ which is close to the total capacitance measured.

ITO Mott-Schottky analysis

The capacity associated to the ITO/PBS interface is measured at different offset potentials: the Mott-Schottky plot is displayed in Fig.(7). For each DC offset the frequency is logarithmically varied from 0.1 Hz up to 10 Hz, in order to sample the impedance only in the capacitive regime. The acquired data are fitted with a linear plot, in order to extract, according to Eq.(4), the dopant

density N_D from the slope and the flat band potential V_{fb} from intercept with the x axis.

The obtained results, with an R^2=0.945, are:

- $V_{fb} = (-0.76 \pm 0.05)\,\mathrm{V}$
- $N_D = (7.5 \pm 0.8) \times 10^{21}\,\mathrm{cm}^{-3}$

The dopant density is in agreement with values found in literature [9].

The voltage range for the Mott-Schottky plot, at which no reactions occur, was determined using cyclovoltammetry as shown in Fig.(8).

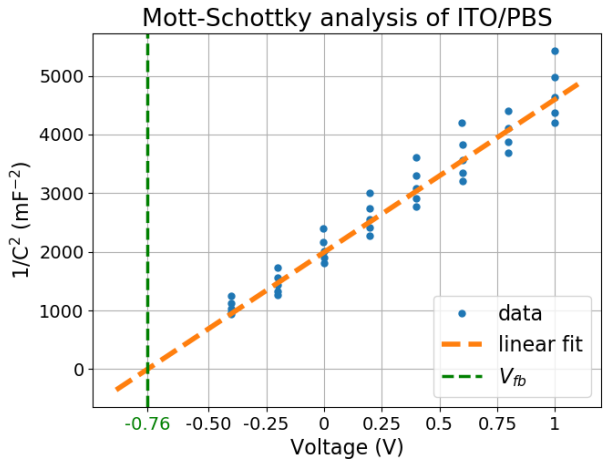


Figure 7: Mott-Schottky plot of the ITO/PBS interface

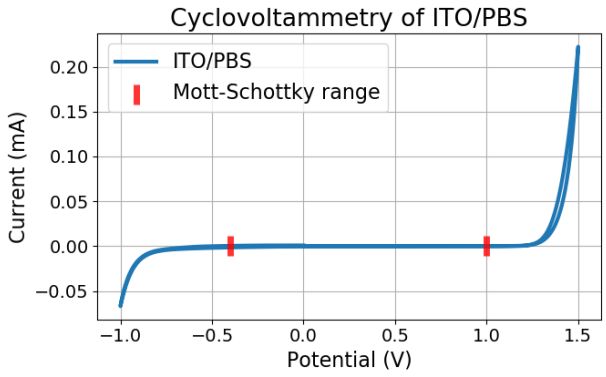


Figure 8: Cyclovoltammetry measure for ITO/PBS interface. The region delimited by the red bars indicates the potential range considered in the Mott-Schottky analysis

PEDOT:PSS layer thickness estimation

The I-V curve related to the electrodeposition of PEDOT:PSS on ITO, obtained during cyclovoltammetry, is shown in Fig.(9). The rise in the amount of current is related with the polymerization processes.

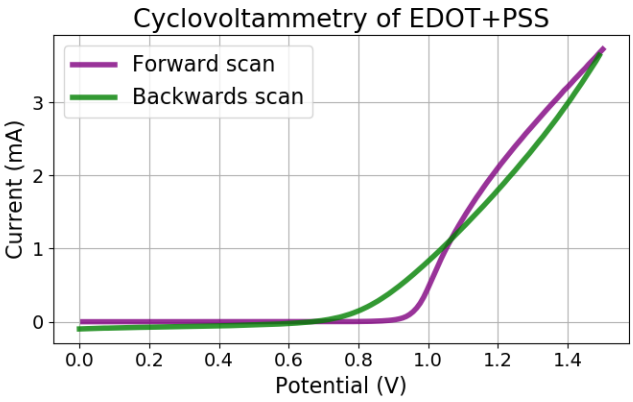


Figure 9: Cyclovoltammetry performed during the electrodepostion of PEDOT:PSS

Even though PEDOT is being formed no particularly prominent oxidation peak is seen indicating that mass transport limitations aren't an issue for a single cyclovoltammetry cycle and that the solution is adequately stirred. While the lack of a reduction peak indicates that the de-polymerization process is kinetically hindered. The small hysteresis in the low potential region is linked to the fact that the capacity of the interface has increased and with it also the constant capacitive current ($I = dQ/dt = C \cdot dV/dt$).

The thickness of the PEDOT:PSS layer can be estimated by determining the total used charge Q, given by the integral over time of the current (Fig.(10)) as for each electron that passed trough the electrode a single EDOT molecule is deposited: the total charge is directly linked to the moles of PEDOT stuck to the electrode. Using Eq.(1), the layer volume can be expressed, without considering any specific geometry, as:

$$V = \frac{MQ}{eN_A\rho}$$

where M is the molecular weight of EDOT (142.17 g/mol) and ρ is the EDOT density (1.34 g/cm^3 [3]).

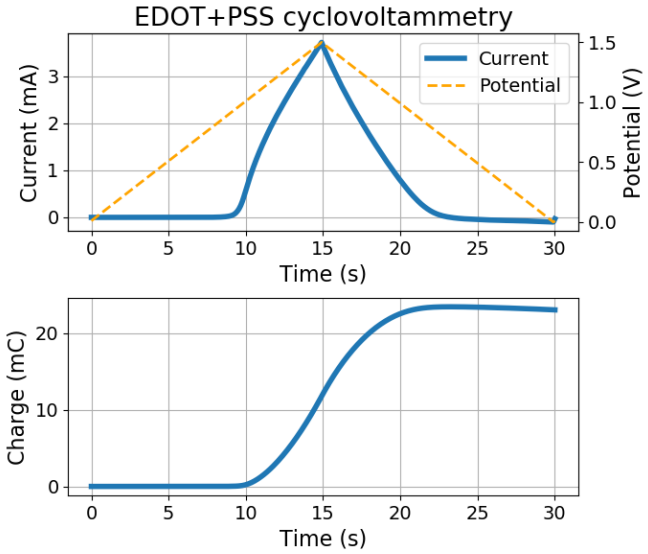


Figure 10: Current/potential and total charge charge time profiles during the electrodeposition

The calculated volume is 25.2 nL (0.0252 mm^3) deposited on a surface are of 1.5 cm^2 giving a layer thickness of $l = (168 \pm 8)$ nm.

By multiplying the capacitance per unit volume found in literature (39 F/cm^2) by the calculated volume, a value of 990 ± 50 µF for the volumetric capacitance of PEDOT:PSS is obtained, which is close to the one resulting from the fit of the impedance spectroscopy data (831 ± 11 µF).

It can be concluded that the volume calculated with this method allows to obtain reasonable results without making assumptions about any specific geometry or taking into account surface roughness.

Conclusions

PEDOT:PSS is an efficient organic semiconductor which supports both electronic and ionic transport, characterized by a high volumetric capacitance and a remarkable biocompatibility.

In this work, a PEDOT:PSS coated ITO electrode is prepared by electrodeposition and is subsequently electrically characterized. During the electrodeposition process, a cyclovoltammetry measurement is performed in order to determine a layer thickness of (168 ± 8) nm and a capacitance C_V of 987 µF.

ITO/PBS and ITO/PEDOT:PSS/Water interfaces are studied by means of impedance spectroscopy. In both cases, the acquired data are fitted with a RC series circuit and the following results are obtained: $R_{bare} = (30.8 \pm 0.9)\,\Omega$ and $C_{bare} = (18.5 \pm 0.5)$ µF for the bare ITO electrode and $R_{coat} = (27.3 \pm 0.2)\,\Omega$ and $C_V = (831 \pm 11)$ µF for the PEDOT:PSS coated electrode. The introduction of PEDOT:PSS leads to an increase of the system capacity of nearly two order of magnitude, while keeping the conductivity high.

The bare ITO electrode is also studied by means of a Mott-Schottky analysis, which provides a dopant density of $N_D = (7.5 \pm 0.8) \times 10^{21}\,cm^{-3}$ and a flat band potential of $V_{fb} = (-0.76 \pm 0.05)$ V.

To further the characterization of the ITO/PEDOT:PSS/Water interface a Mott-Schottky analysis could be helpful in order to compare it to the bare ITO M-S analysis. Another experiment that could be carried out is the electrical and/or optical characterization comparing the doped and undoped PEDOT:PSS states. The frequency response at DC offsets different than 0 V could also be explored.

References

[1] Bryan D. Paulsen et al. "Organic mixed ionic–electronic conductors". In: *Nature Materials* 19 (2020), pp. 13–26. DOI: https://doi.org/10.1038/s41563-019-0435-z.

[2] Lin Zhang, Ke Yang, and Rui Chen. "The Role of Mineral Acid Doping of PEDOT:PSS and Its Application in Organic Photovoltaics". In: *Advanced Electronic Materials* 6.1 (2020). DOI: https://doi.org/10.1002/aelm.201900648.

[3] Elschner Andreas, Kirchmeyer Stephan, and Lovenich Wilfried. *PEDOT: Principles and Applications of an Intrinsically Conductive Polymer*. CRC Press., 2010, p. 51. ISBN: ISBN 978-1-4200-6912-9.

[4] Yu-Cheng Lin, Liang-Yu Chen, and Fu-Chien Chiu. "Lossy Mode Resonance-Based Glucose Sensor with High-k Dielectric Film". In: *Crystals* 9.450 (2019). DOI: 10.3390/cryst9090450.

[5] Antonio J. Olivares et al. "Nanostructural Modification of PEDOT:PSS for High Charge Carrier Collection in Hybrid Frontal Interface of Solar Cells". In: *Polymers* 11.6 (2019). DOI: 10.3390/polym11061034.

[6] Anton V. Volkov et al. "Understanding the Capacitance of PEDOT:PSS". In: *Adv. Funct. Mater.* 27 (2017). DOI: 10.1002/adfm.201700329.

[7] *Autolab PGSTAT 204 potentiostat*. URL: https://www.metrohm.com/en/products/a/ut20/aut204_s.html.

[8] Weiyang Yang et al. "Characteristics of Transparent, PEDOT:PSS Coated Indium-Tin- Oxide (ITO) Microelectrodes". In: *IEEE Trans Nanotechnol.* 17.4 (2018), pp. 701–704. DOI: 10.1109/TNANO.2017.2785627.

[9] Hamid Reza Fallaha, Mohsen Ghasemia, and Ali Hassanzadehb. "Influence of heat treatment on structural, electrical, impedance and optical properties of nanocrystalline ITO films grown on glass at room temperature prepared by electron beam evaporation". In: *Physica E* 39.1 (2007), pp. 69–74. DOI: https://doi.org/10.1016/j.physe.2007.01.003.